

Milestone – 4

SalesGenie AI – AI Sales
Assistant & Lead Intelligence
Platform

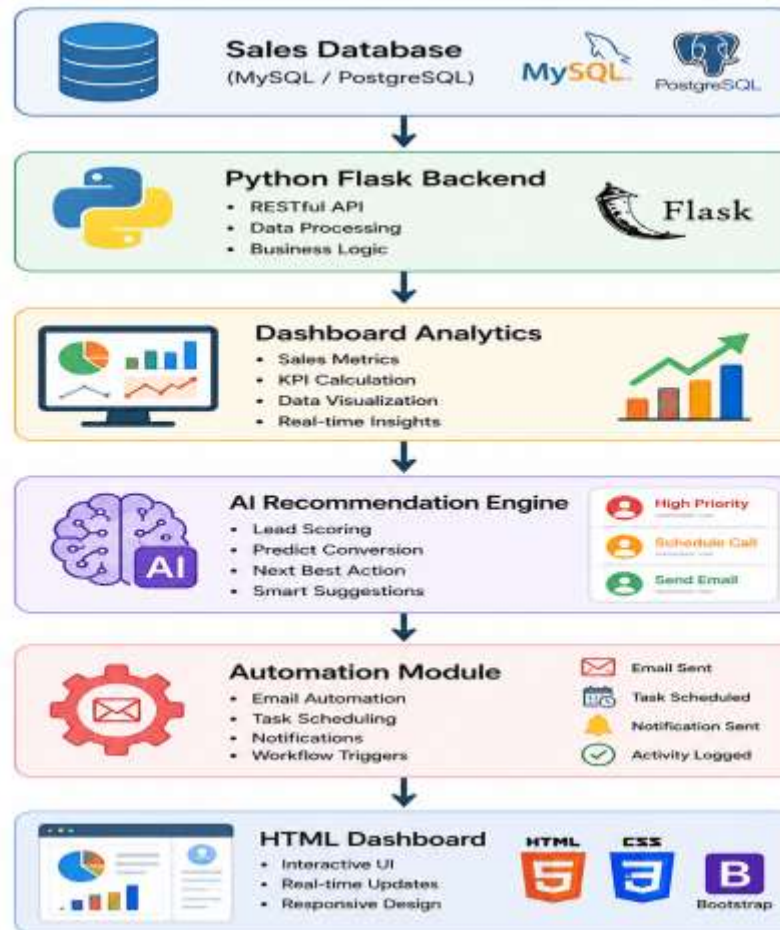
Milestone 4 (Weeks 7–8) – Dashboard & Automation

- Develop sales analytics dashboard.
- Add automated follow-up recommendations.
- Perform testing, optimization, and deployment.

Dashboard & Automation – Explanation with Python Coding

- **Overview**
- The Dashboard is the main interface of the SalesGenie AI platform. It collects data from the database, processes it using the Flask backend, analyzes lead information with AI, and displays interactive sales metrics to the user.

Workflow



1. Conversion Rate Card

Explanation

The Conversion Rate shows the percentage of leads that have been successfully converted into customers.

Formula

$$\text{Conversion Rate} = \frac{\text{Won Leads}}{\text{Total Leads}} \times 100$$

From the dashboard

Conversion Rate



24.8%

This means approximately **25 out of every 100 leads** become customers.

Python Code

```
import sqlite3
```

```
conn = sqlite3.connect("sales.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("SELECT COUNT(*) FROM leads")
```

```
total = cursor.fetchone()[0]
```

```
cursor.execute("SELECT COUNT(*) FROM leads WHERE status='Won'")
```

```
won = cursor.fetchone()[0]
```

```
conversion = (won / total) * 100
```

```
print("Conversion Rate =", round(conversion,2), "%")
```

Output

Conversion Rate = 24.8 %

2. Pipeline Value Card

Explanation

- Pipeline Value represents the total expected revenue from all active opportunities.

Dashboard

Pipeline Value

\$2.4 Million

- It helps managers estimate future sales revenue.

Python Code

```
cursor.execute("SELECT SUM(value) FROM  
leads")
```

```
pipeline_value = cursor.fetchone()[0]  
print("Pipeline Value = $", pipeline_value)
```

Output

Pipeline Value \$2400000

3. Average Response Time

Explanation

- This KPI measures how quickly sales representatives respond to new leads.
- Dashboard
- Average Response Time 2.4 Hours
- Lower response time generally improves customer engagement.

Python Code

```
cursor.execute("SELECT AVG(response_time)
FROM leads")

response = cursor.fetchone()[0]

print("Average Response =",
      round(response,2),"Hours")
```

Output

- Average Response 2.4 Hours

4. Average Sales Cycle

Explanation

- Average Sales Cycle indicates the average number of days required to convert a lead into a customer.
- Dashboard
- 28 Days
- A shorter sales cycle generally means a more efficient sales process.

Python Code

```
cursor.execute("""  
SELECT AVG(sales_cycle)  
FROM leads """)  
cycle = cursor.fetchone()[0]  
print(cycle)
```

5. Sales Pipeline

Explanation

- The Sales Pipeline tracks every lead as it progresses through different stages of the sales process.
- Pipeline
- New Leads Qualified Proposal Negotiation Closed Won
- Each stage shows the companies currently in that phase.

Python Code

```
cursor.execute("""  
SELECT stage,  
COUNT(*)  
FROM leads GROUP BY stage """)  
rows = cursor.fetchall()  
for stage,count in rows:  
    print(stage,count)
```

Output

New Lead 12

Qualified 8

Proposal 5

Negotiation 3

Closed Won 6

6. Dashboard Filter

Explanation

- The dashboard allows the sales manager to view reports for different time periods.

- Options

This Month

This Quarter

This Year

Python Code

```
period = "Month"
if period=="Month":
    query="SELECT * FROM leads WHERE
month='July'"
elif period=="Quarter":
    query="SELECT * FROM leads"
else:
    query="SELECT * FROM leads"
cursor.execute(query)
```

7. AI Follow-up Recommendation

Explanation

- The AI Recommendation Engine analyzes lead activity and suggests the next best action.

Example shown in the dashboard

Follow up with TechCorp

High Priority

The AI considers:

- Days since last contact
- Lead score
- Email activity
- Meeting history
- Proposal status

Python Code

```
def recommendation(days):  
    if days>10:  
        return "High Priority Follow-up"  
    elif days>5:  
        return "Schedule Phone Call"  
    else:  
        return "Send Reminder Email"  
print(recommendation(11))
```

Output

- High Priority Follow-up

8. AI Powered Badge

Explanation

- The **AI Powered** badge indicates that recommendations are generated automatically by the machine learning engine rather than entered manually.
- The AI model can analyze:
 - Customer engagement
 - Email open rate
 - Response history
 - Purchase probability
 - Lead score
- **Example**
 - Lead Score = 91
 - Recommendation
 - Schedule Product Demo

9. Flask Backend

Explanation

- Flask retrieves data from the database and sends it to the dashboard.

Python Code

```
from flask import Flask,render_template
import sqlite3
app=Flask(__name__)
@app.route("/")
def dashboard():
    conn=sqlite3.connect("sales.db")
    cursor=conn.cursor()
    cursor.execute("SELECT COUNT(*) FROM leads")
    total=cursor.fetchone()[0]
    return render_template("dashboard.html", total=total)
app.run(debug=True)
```

10. Dashboard HTML

Explanation

- The frontend displays KPIs using Bootstrap cards.

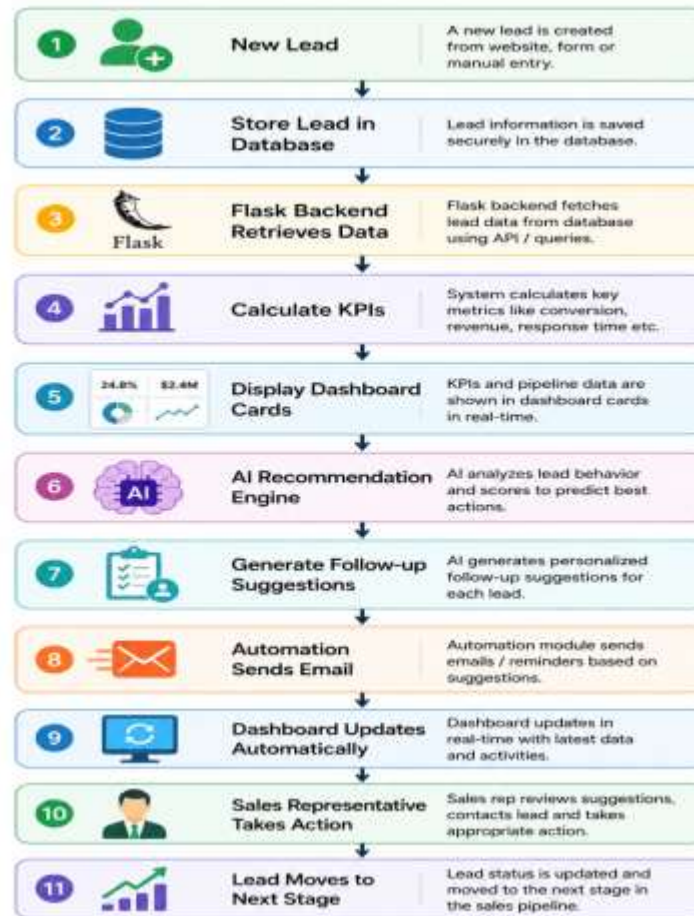
HTML Code

```
<div class="card">  
<h4>Conversion Rate</h4>  
<h2>{{conversion}}%</h2>  
</div> <div class="card">  
<h4>Pipeline Value</h4>  
<h2>${{revenue}}</h2>  
</div>
```

- **11. Dashboard CSS**

```
.card{  
background:white;  
padding:20px;  
border-radius:12px;  
box-shadow:0px 0px 8px gray;  
text-align:center;  
margin:10px;  
}
```

Complete Dashboard Workflow

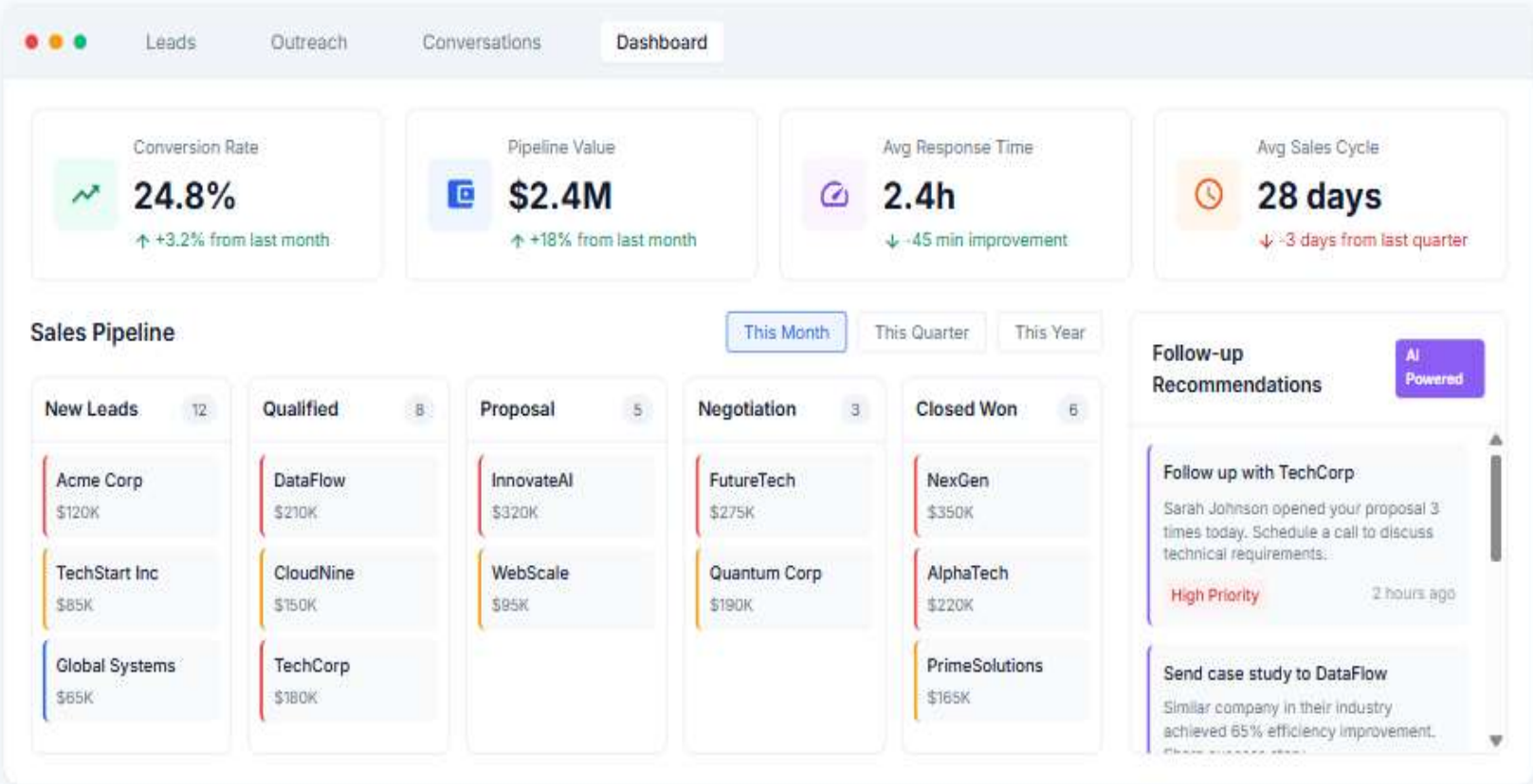


Technologies Used

Module	Technology
Frontend	HTML5, CSS3, Bootstrap, JavaScript
Backend	Python Flask
Database	MySQL / PostgreSQL / SQLite
AI Recommendation	Scikit-learn
Data Processing	Pandas, NumPy
Charts	Matplotlib / Plotly / Chart.js
Automation	schedule, SMTP/SendGrid
Deployment	Gunicorn, Nginx, AWS, Azure, Render

Dashboard & Automation

Sales analytics dashboard with automated follow-up recommendations

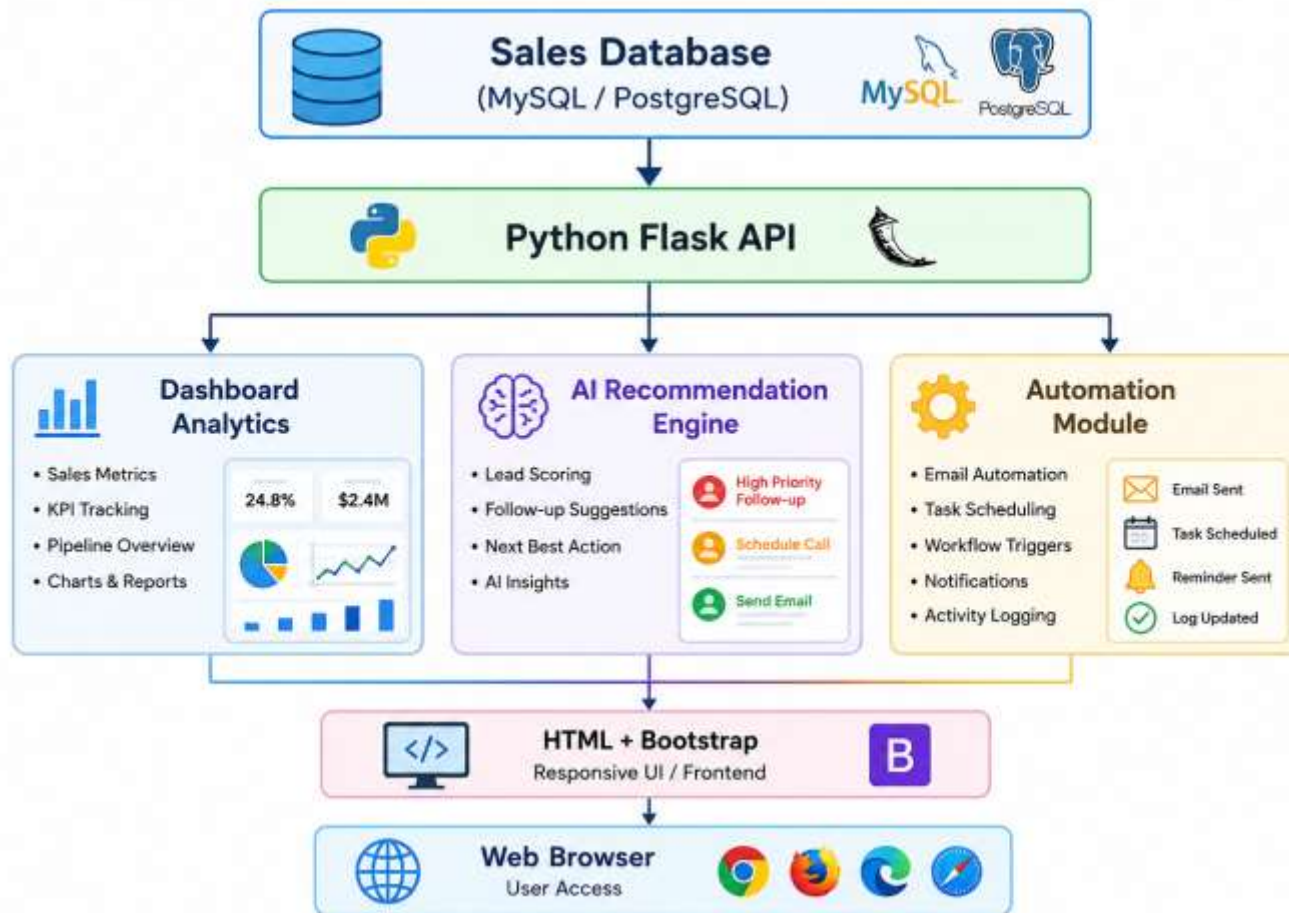


SalesGenie AI – AI Sales Assistant & Lead Intelligence Platform

Objective

- The main objective of Milestone 4 is to build an **AI-powered Sales Dashboard** that enables sales teams to monitor lead performance, analyze business metrics, receive AI-generated follow-up recommendations, and automate repetitive sales activities.
- This dashboard helps sales managers make faster, data-driven decisions while increasing sales productivity.

System Architecture



Module 1: Sales Database

- **Purpose**
- The Sales Database stores all information related to customers, prospects, sales activities, and follow-up history.
- **Responsibilities**
- Store lead details
- Maintain customer information
- Save sales pipeline status
- Record AI recommendations
- Store communication history

Example Database Table

Lead ID	Company	Stage	Revenue	Status
101	Acme Corp	Qualified	\$120K	Open
102	TechStart	Proposal	\$85K	Open

Python Code

```
import sqlite3
conn = sqlite3.connect("sales.db")
cursor = conn.cursor()
cursor.execute("""
CREATE TABLE IF NOT EXISTS leads(
id INTEGER PRIMARY KEY,
company TEXT,

stage TEXT,

value INTEGER,

status TEXT

)

""")

conn.commit()

conn.close()
```

Explanation

- `sqlite3.connect()` creates or opens the database.
- `CREATE TABLE` creates the leads table if it does not already exist.
- The table stores company details, sales stage, revenue value, and lead status.
- `commit()` saves the changes permanently.

Module 2: Python Flask API

Purpose

- Flask acts as the backend server that connects the dashboard to the database.

Responsibilities

- Retrieve sales data
- Calculate KPIs
- Send data to the frontend
- Handle user requests

Python Code

```
from flask import Flask, render_template
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def dashboard():
```

```
    return render_template("dashboard.html")
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

Explanation

- `Flask()` creates the web application.
- `@app.route("/")` defines the homepage.
- `render_template()` loads the dashboard webpage.
- `app.run()` starts the Flask server.

Module 3: Dashboard Analytics

Purpose

- Dashboard Analytics displays real-time business performance using charts and KPI cards.

Dashboard KPIs

- Conversion Rate
- Pipeline Value
- Average Response Time
- Average Sales Cycle
- Total Revenue
- Lead Status

Python Code

```
import sqlite3
```

```
conn=sqlite3.connect("sales.db")
```

```
cursor=conn.cursor()
```

```
cursor.execute("SELECT COUNT(*) FROM leads")
```

```
total=cursor.fetchone()[0]
```

```
cursor.execute("SELECT COUNT(*) FROM leads WHERE status='Won'")
```

```
won=cursor.fetchone()[0]
```

```
conversion=(won/total)*100
```

```
print(conversion)
```

Explanation

- Counts total leads.
- Counts successfully converted leads.
- Calculates conversion rate using:

$$\text{Conversion Rate} = \frac{\text{Won Leads}}{\text{Total Leads}} \times 100$$

This value is displayed on the dashboard.

Module 4: Sales Pipeline

- **Purpose**
- Shows how leads move through each stage of the sales process.
- **Pipeline Stages**

New Lead



Qualified



Proposal



Negotiation



Closed Won

Python Code

```
cursor.execute("""
```

```
SELECT stage,
```

```
COUNT(*)
```

```
FROM leads
```

```
GROUP BY stage
```

```
""")
```

```
pipeline=cursor.fetchall()
```

```
print(pipeline)
```


Explanation

The SQL query groups all leads based on their current sales stage.

Example Output

New Lead : 15

Qualified : 10

Proposal : 8

Negotiation : 4

Closed Won : 2

This information is displayed in the dashboard pipeline.

Module 5: AI Recommendation Engine

Purpose

- The AI Recommendation Engine suggests the next best action for every lead based on engagement history and inactivity.

Decision Logic



Python Code

```
def followup(days):
```

```
    if days>10:
```

```
        return "High Priority Follow-up"
```

```
    elif days>5:
```

```
        return "Schedule Phone Call"
```

```
    else:
```

```
        return "Send Reminder Email"
```

Explanation

- If the lead has not been contacted for more than 10 days, the system marks it as **High Priority**.
- If there has been no contact for 6–10 days, it recommends a phone call.
- Otherwise, it sends a reminder email.
- This helps sales representatives prioritize important leads.

Module 6: Automation Module

Purpose

- Automatically performs repetitive sales tasks.

Automated Activities

- Email reminders
- Meeting scheduling
- CRM updates
- Daily reports
- Notifications

Python Code

```
import schedule
```

```
import time
```

```
def send_email():
```

```
    print("Follow-up Email Sent")
```

```
schedule.every().day.at("10:00").do(send_email)
```

```
while True:
```

```
    schedule.run_pending()
```

```
    time.sleep(1)
```

Explanation

- The schedule library automates recurring tasks.
- Every day at 10:00 AM, the application executes the `send_email()` function.
- This reduces manual effort and ensures timely follow-ups.

Module 7: Dashboard Frontend

- **Purpose**
- The frontend displays analytics received from the Flask backend.

HTML Example

```
<div class="card">  
<h3>Conversion Rate</h3>  
<h1>24.8%</h1>  
</div>
```

Explanation

- Each KPI is shown as a card containing:
- KPI title
- Current value
- Trend indicator (increase/decrease)
- This provides a clear summary of business performance.

Module 8: Testing

Purpose

- Testing verifies that every module works correctly.

Unit Test

```
assert followup(12)=="High Priority Follow-up"  
print("Test Passed")
```

Explanation

- The test checks whether the AI recommendation function returns the expected result for a lead inactive for 12 days. If the output matches, the test passes.

Module 9: Optimization

- **Purpose**
- Improve application performance and responsiveness.
- **Optimization Techniques**
- Create database indexes for faster searches.
- Cache frequently accessed dashboard metrics.
- Reduce unnecessary SQL queries.
- Load charts asynchronously.
- Retrain AI models periodically for improved prediction accuracy.

Module 10: Deployment

- **Purpose**
- Deploy the SalesGenie AI application so users can access it through a web browser.

Installation

pip install flask

pip install pandas

pip install matplotlib

pip install scikit-learn

Run Application

- python app.py

Access the Dashboard

- <http://127.0.0.1:5000>

Complete Dashboard Workflow



Advantages of Milestone 4

- Provides real-time visibility into sales performance.
- Enables data-driven decision-making with KPIs and analytics.
- Uses AI to recommend the next best follow-up action.
- Reduces manual work through automation.
- Improves response time and lead conversion rates.
- Supports deployment as a scalable web application using Python Flask.

This milestone transforms **SalesGenie AI** from a simple lead management system into an intelligent **AI-powered Sales Analytics and Automation Platform**, enabling organizations to monitor sales performance, automate routine tasks, and prioritize high-value leads efficiently.

